COMMON BASELINE VALUES

AND STANDARD FORMAT FOR ALL PLATFORMS

ATU VIRTUAL LAB AND AUTOMOTIVE SIMULATION PLATFORM

Prepared For Lecturer And Official Review

Engineering Faculty - Mechanical / Automotive Engineering Focus

Current Scope

This file defines the common vehicle identity, SI unit standard, signal naming style, and shared vehicle-dynamics meaning for integration.

BASELINE STANDARDS

Common Baseline Values and Standard Format for All Platforms

This document defines the common baseline values and standard formatting rules to be used across all platform models. The purpose is not to restrict each model, but to make sure that all models begin from the same vehicle identity, use the same measurement system, and produce results that can be integrated later.

The common vehicle identity selected for the project is the Genesis GV70. All platform models should use the baseline values below unless a specific model clearly states that it is changing a value for testing, comparison, sensitivity analysis, or a topic-specific requirement.

Table 1: Standards Used Across the Project

Standard Area	Standard Used	Plain Meaning / Application
Metric / SI Unit Standard	BIPM International System of Units (SI)	All platform models should use metric/SI units for calculations, tables and results.
Vehicle Identity Standard	Genesis GV70 baseline vehicle	All models begin from the same vehicle body identity so shared results can be compared and integrated.
Vehicle Dynamics Meaning Standard	ISO 8855 road vehicle dynamics vocabulary	Terms such as yaw, pitch, roll, wheel slip, understeer and oversteer should have the same meaning across the project.
Signal Naming Standard	Project-defined System_Parameter naming style	Common signals should use clear names with underscores, such as Vehicle_Speed and Battery_SOC.
Adjustable Scenario Rule	Project-defined modelling rule	Final drive, axle layout, road gradient, friction, payload, drive cycle and component sizes should remain adjustable.

Table 2: Common Vehicle Baseline Values

Item	Standard Value	Unit	Naming Style	Plain Meaning / Use
Vehicle Identity	Genesis GV70	-	Vehicle_ID	Common reference vehicle for all models.
Vehicle Mass	2000	kg	Vehicle_Mass	Rounded default modelling value for motion, braking, suspension, energy and stability models.
Number of Tyres/Wheels	4	-	Number_of_Wheels	Common wheel quantity for wheel-speed, braking, suspension and stability models.
Tyre Size Reference	235/55R19	-	Tyre_Size	Reference tyre size for estimating the wheel radius.
Tyre/Wheel Radius	0.371	m	Wheel_Radius	Connects vehicle speed, wheel speed, motor speed and torque conversion.
Wheelbase	2.875	m	Wheelbase	Distance between front and rear wheel centres.
Front Track Width	1.649	m	Front_Track_Width	Distance between left and right front wheels.
Rear Track Width	1.665	m	Rear_Track_Width	Distance between left and right rear wheels.
Average Track Width	1.657	m	Track_Width	Average track width for handling, load-transfer and stability models.
Centre of Gravity Height	0.600	m	CG_Height	Assumed default modelling value for braking, cornering and load-transfer analysis.

Item	Standard Value	Unit	Naming Style	Plain Meaning / Use
Overall Length	4.715	m	Vehicle_Length	Reference body length only; not a compulsory model input for every topic.
Overall Width	1.910	m	Vehicle_Width	Reference body width only; not a compulsory model input for every topic.
Overall Height	1.630	m	Vehicle_Height	Reference body height for vehicle identity and model description.

Note: The vehicle mass is rounded to 2000 kg for modelling simplicity. Exact GV70 mass varies by trim and market. The CG height is an assumed default modelling value and may be varied only when a model is performing sensitivity analysis or load-transfer comparison.

Table 3: Metric / SI Unit Standard

Quantity	Standard Unit	Symbol	Example Naming Style	Plain Meaning
Mass	kilogram	kg	Vehicle_Mass	Mass-related modelling should be entered in kg.
Length / Distance	metre	m	Wheelbase	All physical dimensions should be entered in metres.
Time	second	s	Simulation_Time	Time-based simulation should use seconds.
Speed	metre per second	m/s	Vehicle_Speed	Main calculation speed.
Display Speed	kilometre per hour	km/h	Vehicle_Speed_kmh	Speed may be displayed in km/h for easy understanding.
Acceleration	metre per second squared	m/s^2	Vehicle_Acceleration	Used for motion, braking and stability.
Force	newton	N	Brake_Force	Used for braking, tyre, suspension and load forces.
Torque	newton metre	Nm	Motor_Torque	Used for engine, motor, wheel and braking torque.
Power	kilowatt	kW	Motor_Power	Used for engine, motor, inverter and battery power.
Energy	kilowatt-hour / joule	kWh / J	Battery_Energy	kWh for battery energy; J for physics calculations.
Voltage	volt	V	Battery_Voltage	Electrical potential.
Current	ampere	A	Battery_Current	Electrical current flow.
Temperature	degree Celsius	deg C	Battery_Temperature	Temperature display and thermal monitoring.
Rotational Speed	revolutions per minute	rpm	Engine_Speed	Used for engine, motor and wheel rotation.
Angle	degree	deg	Steering_Angle	Used for steering, yaw, pitch and roll display.
Pressure	pascal / bar	Pa / bar	Brake_Pressure	Pa for calculation; bar may be shown for practical brake pressure.

Table 4: Standard Signal Naming Style

Rule	Format	Example	Plain Meaning
Use clear engineering names	Words separated by underscore	Vehicle_Speed	Easy to read and easy to connect between models.
Start with system name	System + parameter	Battery_SOC	Shows which system the signal belongs to.
Avoid unclear short names	Do not use vague labels	Use Vehicle_Speed, not VS	Prevents confusion during integration.
Use same name across platforms	Same signal, same label	Wheel_Speed	Wheel speed should not be renamed differently by each person.
Keep units separate	Unit recorded in table, not inside signal name	Vehicle_Speed, unit = m/s	Keeps naming clean and easy to export.
Add display version only when needed	Use suffix for display	Vehicle_Speed_kmh	Calculation remains SI while display can be user-friendly.

Table 5: Core Signal Names for Model Integration

Signal Name	Unit	Used By	Plain Meaning
Vehicle_Speed	m/s	BEV, HEV, ICE, vehicle dynamics	Forward speed of the vehicle.
Vehicle_Speed_kmh	km/h	Display/output	Same speed shown in km/h.
Vehicle_Acceleration	m/s^2	Motion, braking, powertrain	Rate of change of vehicle speed.
Wheel_Speed	rpm or rad/s	Motor drive, ABS, stability	Rotational speed of the wheel.
Wheel_Torque	Nm	Motor, engine, braking	Torque delivered to or resisted at the wheel.
Steering_Angle	deg	Stability, steering, handling	Driver steering input.
Brake_Force	N	Braking, ABS, stability	Braking force applied.
Brake_Pressure	Pa or bar	Braking, ABS model	Hydraulic or equivalent brake pressure.
Yaw_Rate	deg/s or rad/s	Stability control	Vehicle rotation around vertical axis.
Roll_Angle	deg	Suspension, handling	Body tilt from side to side.
Pitch_Angle	deg	Suspension, braking	Body tilt forward or backward.
Suspension_Displacement	m	Suspension, shock model	Shock or suspension travel.
Battery_SOC	%	BEV, HEV	Battery state of charge.
Battery_Voltage	V	BEV, HEV, power electronics	Battery electrical voltage.
Battery_Current	A	BEV, HEV, power electronics	Battery current.
Battery_Temperature	deg C	Battery thermal model	Battery temperature.
Motor_Speed	rpm	BEV, HEV	Electric motor speed.
Motor_Torque	Nm	BEV, HEV	Electric motor torque.
Motor_Power	kW	BEV, HEV	Electric motor power.
Engine_Speed	rpm	ICE, HEV	Engine rotational speed.
Engine_Torque	Nm	ICE, HEV	Engine output torque.
Fuel_Rate	L/h or g/s	ICE, HEV	Fuel consumption rate.

Table 6: Vehicle Dynamics Meaning Standard

This section keeps the language of vehicle dynamics consistent. It does not fix the testing method; it only ensures that the words used in the different platform models mean the same thing.

Term	Plain Meaning	Use in the Project
Yaw	Vehicle turning around its vertical axis.	Used in stability and cornering models.
Pitch	Vehicle nose moving up or down.	Used in braking, acceleration and suspension models.
Roll	Vehicle body leaning left or right.	Used in cornering and suspension models.
Wheel Slip	Difference between tyre rotation and actual road movement.	Used in ABS, traction, braking and stability models.
Understeer	Vehicle turns less than the driver demands.	Used in handling and stability analysis.
Oversteer	Vehicle turns more than the driver demands.	Used in handling and stability analysis.
Load Transfer	Movement of vertical load between tyres during braking, acceleration or cornering.	Used in braking, suspension and stability models.
Steering Angle	Angle input from the steering system.	Used in steering, handling and stability control.
Suspension Travel	Up-and-down movement of the suspension or shock absorber.	Used in suspension and physical shock model testing.

Plain explanation: When one person says yaw, pitch, roll, wheel slip, understeer or oversteer, everybody should understand the same meaning. ISO 8855 can be referenced by the vehicle-dynamics topics because it defines the main terms used for road vehicle dynamics and road-holding ability.

Table 7: Values That Should Remain Adjustable

Parameter	Reason It Should Not Be Fixed
Final Drive Ratio	Different models may use different drivetrain and reduction arrangements.
Axle Layout	BEV, HEV, ICE and physical models may use different layouts.
Road Gradient	Should be adjustable for uphill, downhill and level-road studies.
Road Friction	Should vary for dry, wet, slippery or loose surfaces.
Drag Coefficient	Can vary depending on aerodynamic assumptions.
Frontal Area	Can vary if model body assumptions change.
Rolling Resistance	Should vary by tyre and road condition.
Payload	Should vary for loading studies.
Drive Cycle	Each person may use different test conditions.
Engine Size	Specific to ICE and HEV models.
Motor Size	Specific to BEV and HEV models.
Battery Capacity	Specific to BEV and HEV models.
Suspension Stiffness	Specific to suspension and handling model.
Damping Coefficient	Specific to shock and suspension model.

Final Rule: All platform models should use Genesis GV70 as the common vehicle identity, metric/SI units throughout, common signal names, and common vehicle-dynamics meanings. Model-specific values may change, but the change must be clearly stated inside that person's work.

Reference Notes

[1] BIPM describes SI as the International System of Units. This project adopts metric/SI units as the common measurement standard.

[2] Genesis GV70 specification information lists overall length as 4,715 mm, overall width as 1,910 mm, overall height as 1,630 mm, front tread as 1,649 mm, rear tread as 1,665 mm and wheelbase as 2,875 mm for the 19-inch tyre package.

[3] Edmunds lists the 2025 Genesis GV70 2.5T curb weight as 4,167 lb; the project uses 2,000 kg as a rounded baseline modelling value for easier group integration.

[4] ISO 8855:2011 defines principal terms used in road vehicle dynamics and road-holding ability. It is useful as a terminology reference for the vehicle dynamics platforms.

[5] The signal naming standard and adjustable scenario rule are project-defined integration rules created for this final-year platform project.